Lecture 2

- In Lecture 1, we defined linear ODEs+IVPs and revised integrating factors

Definition (Linear ODEs)

$$y^{(n)}(t) + \ldots + f_1(t)y'(t) + f_0(t)y(t) = g(t)$$

Definition (Initial value problems, IVPs)

$$\text{Usually } y(t_0) = y_0,\ y'(t_0) = y_0',\ \ldots \text{ and } y^{(n-1)}(t_0) = y_0^{(n-1)}$$

- In Lecture 2, we will:
  1. Establish a connection between 1st-order ODEs and linear algebra
  2. Express ODE solutions in terms of complementary and particular solutions
  3. Extend concepts to 2nd-order ODEs
  4. Learn how to use the reduction of order method

Connection to linear algebra

- Linear algebra: $\mathbf{A}\boldsymbol{v} = \boldsymbol{w}$ where $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{w} \in \mathbb{R}^m$.
- $\mathbf{A} : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation:

$$\mathbf{A}(\boldsymbol{u}+\boldsymbol{v}) = \mathbf{A}\boldsymbol{u} + \mathbf{A}\boldsymbol{v} \quad \text{and} \quad \mathbf{A}(k\boldsymbol{u}) = k\mathbf{A}\boldsymbol{u} \quad \text{where} \quad \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n;\ k \in \mathbb{R}$$

- $\frac{\mathrm{d}}{\mathrm{d}t}$ and $f(t)\cdot$ are also linear transformations but they operate on functions.*
- $\mathsf{T} = \frac{\mathrm{d}}{\mathrm{d}t} + f(t)\cdot$ is a *linear transformation*

$$(\mathsf{T}y)(t) = \frac{\mathrm{d}y}{\mathrm{d}t} + f(t)y(t)$$

...you should check the definition is satisfied, i.e.,

$$\mathsf{T}(x+y) = \mathsf{T}x + \mathsf{T}y \quad \text{and} \quad \mathsf{T}(k\,y) = k\,\mathsf{T}y$$

*In Topic 4 we will extend the idea of these finite linear combinations to an infinite series. If you enjoy this aspect of the course then take MAST30031 Methods of Mathematical Physics and/or MAST30026 Metric & Hilbert Spaces to gain a more detailed understanding.

We have the following connections

$\mathbf{A}\boldsymbol{v} = \boldsymbol{w}$	$\leftrightarrow$	$y'(t) + f(t)y(t) = g(t)$
$\boldsymbol{v}$	$\leftrightarrow$	$y(t)$
$\boldsymbol{w}$	$\leftrightarrow$	$g(t)$
$\mathbf{A}$	$\leftrightarrow$	$\frac{\mathrm{d}}{\mathrm{d}t} + f(t)\cdot$

How to solve $\mathbf{A}\boldsymbol{v} = \boldsymbol{w}$?

- Solve $\mathbf{A}\boldsymbol{v}_c = \mathbf{0}$ to find complementary solutions.
  - Complementary solutions form a vector space, i.e., the solution space.
- Find a single solution to $\mathbf{A}\boldsymbol{v}_p = \boldsymbol{w}$, i.e., a particular solution.
- General solution: $\boldsymbol{v} = \boldsymbol{v}_c + \boldsymbol{v}_p$.

The same procedure works for linear ODEs

To solve $\mathsf{T}y = g$

- Solve $\mathsf{T}y_c = 0$ to find complementary solutions
  - Complementary solutions form a solution space.
- Find a (single) particular solution to $\mathsf{T}y_p = g$
- General solution: $y = y_c + y_p$

Exercise 1.4: Recall previous exercise

We considered the linear, first-order, inhomogeneous ODE, using an IF.

$$y'(t) - \frac{1}{t}y(t) = 1 \quad \Rightarrow \quad y(t) = t\ln t + Ct$$

Find:

- Complementary solutions: $y_c = Ct$ (solution space is one-dimensional)
- Particular solution: $y_p = t\ln t$
- General solution: $y = Ct + t\ln t$

Second order linear ODEs

- Using notation from the linear algebra analogy, consider ODEs of form

$$(\mathsf{T}y)(t) = g(t) \quad \text{where} \quad \mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$$

- To find solutions we will use

Topic 1: Trial solutions ("ansätze"), reduction of order, matrix diagonalisation

Topic 2: Power series

Topic 3: Laplace transforms

Topic 4: Fourier series

Topic 6: Fourier transforms

Trial solutions for constant coefficient ODEs (revision)

Consider $\mathsf{T}y = 0$, where

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + b\frac{\mathrm{d}}{\mathrm{d}t} + c \quad \text{such that} \quad b, c \in \mathbb{R}$$

To solve, recall the approach:

- Trial a solution of the form $y(t) = \mathrm{e}^{\lambda t}$, where $\lambda \in \mathbb{C}$ (this is called an "ansatz").
- This leads to the characteristic equation $\lambda^2 + b\lambda + c = 0$
- Then $\lambda_\pm = \dfrac{-b \pm \sqrt{b^2 - 4c}}{2}$
- Therefore, $y = C_+\mathrm{e}^{\lambda_+ t} + C_-\mathrm{e}^{\lambda_- t}$ if $b^2 - 4c \neq 0$ so that $\lambda_+ \neq \lambda_-$

Reduction of order

Solve $\mathsf{T}y = 0$ for the case where $c = \dfrac{b^2}{4} \Rightarrow b^2 - 4c = 0$

- Method of trial solution only gives one solution, $y(t) = \mathrm{e}^{-\frac{b}{2}t}$
- Trial solution of the form $y(t) = u(t)\mathrm{e}^{-\frac{b}{2}t}$
- $y' = \left(u' - \dfrac{b}{2}u\right)\mathrm{e}^{-\frac{b}{2}t}$ and $y'' = \left(u'' - bu' + \dfrac{b^2}{4}u\right)\mathrm{e}^{-\frac{b}{2}t}$
- $\mathsf{T}y = \left(u'' + u'(-b+b) + u(b^2/4 - b^2/2 + b^2/4)\right)\mathrm{e}^{-\frac{b}{2}t} = u''\mathrm{e}^{-\frac{b}{2}t} = 0$
- $u'' = 0$, so $u = C_1 t + C_2$ and hence we have found the general solution

$$y(t) = (C_1 t + C_2)\,\mathrm{e}^{-\frac{b}{2}t}$$

Reduction of order—general theory

Definition

Let $v(t)$ be a non-zero solution of $\mathsf{T}v = 0$ where T is an arbitrary second-order linear operator, then the general solution to $\mathsf{T}y = g$ can be found in the form $y(t) = u(t)v(t)$.

- Chain rule gives $y' = u'v + uv'$ and $y'' = u''v + 2u'v' + uv''$
- Therefore

$$\begin{aligned}\mathsf{T}y &= \big[u''v + 2u'v' + uv''\big] + f_1\big[u'v + uv'\big] + f_0\big[uv\big] \\ &= u\,\mathsf{T}v + u''v + 2u'v' + f_1u'v\end{aligned}$$

- $\mathsf{T}v = 0$ and hence $\mathsf{T}y = u''v + 2u'v' + f_1u'v = g$. No f_0 (explicitly)
- Rearranging gives $u'' + \left(\dfrac{2v'}{v} + f_1\right)u' = \dfrac{g}{v}$. What do you notice?

- Let $w = u'$, so $w' + \left(\dfrac{2v'}{v} + f_1\right) w = \dfrac{g}{v}$
- We have *reduced* the order of our ODE from two to one.
- Solve using an integrating factor:

$$I(t) = \mathrm{e}^{\int (\frac{2v'}{v} + f_1)\mathrm{d}t} = \mathrm{e}^{2\log v + \int f_1 \mathrm{d}t} = \mathrm{e}^{\log v^2}\, \mathrm{e}^{\int f_1 \mathrm{d}t} = v^2 \mathrm{e}^{\int f_1 \mathrm{d}t}$$

- Thus

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[Iw\right] = \frac{Ig}{v} \quad \Rightarrow \quad w = \frac{1}{I}\int \frac{Ig}{v}\mathrm{d}t$$

$$\Rightarrow \quad u = \int \left[\frac{1}{I}\int \frac{Ig}{v}\mathrm{d}t\right]\mathrm{d}t \quad \Rightarrow \quad y = v\int \left[\frac{1}{I}\int \frac{Ig}{v}\mathrm{d}t\right]\mathrm{d}t$$

Note: the two integrals will give us the required two unknown constants.

Summary of reduction of order

1. Given known complementary solution, $v(t)$, to a second order ODE, write general solution as $y(t) = u(t)v(t)$
2. This reduces the order of the ODE to one: $w' + \left(\frac{2v'}{v} + f_1\right) w = \frac{g}{v}$ where $w = u'$
3. Use the integrating factor $I = v^2 \mathrm{e}^{\int f_1 \mathrm{d}t}$ to solve for w.
4. This leads to the general solution $y = v \int \left[\frac{1}{I} \int \frac{Ig}{v} \mathrm{d}t\right] \mathrm{d}t$

Finding the first solution for $\mathsf{T}v = 0$ is usually the difficult task.

Exercise 1.5: Apply reduction of order to the ODE

$$y'' + \frac{1}{t}y' - \frac{1}{t^2}y = g(t) \quad \text{given that } y = t \text{ is a complementary solution}$$

We write the solution as $y(t) = v(t)\,u(t) = t\,u(t)$.

We therefore have $y' = (t\,u)' = tu' + u$ and $y'' = (t\,u' + u)' = tu'' + 2u'$.

Reduction of order gives

$$w' + \frac{3}{t}w = \frac{g}{t} \quad \text{where} \quad w = u'$$

Integrating factor is $I(t) = v^2 \mathrm{e}^{\int t^{-1}\,\mathrm{d}t} = t^2 \mathrm{e}^{\ln(t)} = t^2 \times t = t^3 \quad \Rightarrow \quad w = \dfrac{1}{t^3}\displaystyle\int t^2\, g\,\mathrm{d}t$

Find the general solution when $g = 0$

$$t^3 w = \int t^2 \cdot 0 \, \mathrm{d}t = C_1 \qquad \Rightarrow \qquad w = u' = \frac{C_1}{t^3} \qquad \Rightarrow \qquad u = -\frac{C_1}{2t^2} + C_2$$

Therefore

$$y = t\,u = -\frac{C_1}{2t} + C_2 t$$

Find the general solution when $g = \mathrm{e}^t$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(t^3 w\right) = t^2\mathrm{e}^t$$

$$t^3 w = \int t^2\mathrm{e}^t\mathrm{d}t = t^2\mathrm{e}^t - \int 2t\mathrm{e}^t\mathrm{d}t + C_1$$

$$t^3 w = t^2\mathrm{e}^t - 2t\mathrm{e}^t + \int 2\mathrm{e}^t\mathrm{d}t + C_1 = (t^2 - 2t + 2)\mathrm{e}^t + C_1$$

$$(w =)u' = \frac{\mathrm{e}^t}{t} - \frac{2\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t^3} + \frac{C_1}{t^3}$$

$$u = \int \left\{\frac{\mathrm{e}^t}{t} - \frac{2\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t^3}\right\}\mathrm{d}t - \frac{C_1}{2t^2} + C_2$$

$$= \frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t}{t^2} - \frac{C_1}{2t^2} + C_2$$

$$y = t\left(\frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t}{t^2} - \frac{C_1}{2t^2} + C_2\right) = \left(1 - \frac{1}{t}\right)\mathrm{e}^t - \frac{C_1}{2t} + C_2 t$$

Workings

$$\begin{aligned}\int \left\{ \frac{\mathrm{e}^t}{t} - \frac{2\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t^3} \right\} \mathrm{d}t &= \int \frac{\mathrm{e}^t}{t} \mathrm{d}t - 2\left(-\frac{\mathrm{e}^t}{t} + \int \frac{\mathrm{e}^t}{t} \mathrm{d}t \right) + 2\left(-\frac{\mathrm{e}^t}{2t^2} + \int \frac{\mathrm{e}^t}{2t^2} \mathrm{d}t \right) \\ &= -\frac{\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t} - \int \frac{\mathrm{e}^t}{t} \mathrm{d}t + \left(-\frac{\mathrm{e}^t}{t} + \int \frac{\mathrm{e}^t}{t} \mathrm{d}t \right) \\ &= \frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t}{t^2}\end{aligned}$$